

MATH 301 - Applied Linear Algebra

Professor: Dr. Sarah Chen | schen@university.edu | Science Hall 412

Office Hours: Mon & Wed 11:00 AM - 12:30 PM, or by appointment

Term: Fall 2026 | Credits: 3

Location: See schedule below

Course Description

This course provides a rigorous yet applied treatment of linear algebra, covering vector spaces, matrix theory, and linear transformations with an emphasis on computational applications. Students will implement algorithms in Python (NumPy/SciPy) and apply linear algebra to problems in data science, graphics, and optimization. Topics include Gaussian elimination, LU and QR factorization, eigenvalue decomposition, the Singular Value Decomposition (SVD), and Principal Component Analysis (PCA). The course meets Monday, Wednesday, and Friday from 10:00-10:50 AM in Science Hall 201.

Learning Objectives

By the end of the course, students will be able to:

1. Solve systems of linear equations using direct and iterative methods.
2. Compute and interpret eigenvalues and eigenvectors.
3. Apply the SVD to dimensionality reduction and least-squares problems.
4. Implement matrix algorithms in Python with NumPy.
5. Analyze the numerical stability of linear algebra computations.

Required Materials

Textbook: 'Introduction to Linear Algebra' by Gilbert Strang, 5th ed. (MIT Press).

Software: Python 3.11+ with NumPy, SciPy, Matplotlib (Anaconda distribution recommended).

A scientific calculator is required for exams.

Assignments & Due Dates

Homework 1 - Vectors, Matrices, and Linear Systems - Due: September 18, 2026 (10%)

Problems 1-30 from Chapter 1 & 2. Topics: vector operations, matrix multiplication, Gaussian elimination, and row echelon form. Submit a PDF of hand-written or typed solutions on Gradescope by 11:59 PM.

Homework 2 - Linear Independence, Bases, and Rank - Due: October 2, 2026 (10%)

Problems from Chapters 3 & 4. Topics: null space, column space, four fundamental subspaces, rank-nullity theorem. Submit on Gradescope.

Midterm Exam - Due: October 16, 2026 (20%)

In-class exam covering Chapters 1-4: linear systems, matrix factorizations, subspaces, and orthogonality. Closed-book; one 3x5 index card of notes allowed. Held during regular class time in Science Hall 201.

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Homework 3 - Eigenvalues and Eigenvectors - Due: October 30, 2026 (10%)

Problems from Chapter 5. Topics: characteristic polynomial, diagonalization, symmetric matrices, and the spectral theorem. Python portion: compute eigenvalues of a 5x5 matrix using both NumPy and the power iteration method.

Homework 4 - SVD and Least Squares - Due: November 13, 2026 (10%)

Problems from Chapters 6 & 7. Topics: Singular Value Decomposition, pseudoinverse, least-squares regression, and low-rank approximation. Python coding problems included.

Computational Project - PCA on Real Data - Due: December 4, 2026 (20%)

Apply PCA to a dataset of your choosing (minimum 500 samples, 10+ features). Submit a Jupyter notebook (.ipynb) and a 4-page report covering: data preprocessing, explained variance analysis, 2D/3D visualisation, and interpretation. Groups of 1-2 students. Submit via course portal by 11:59 PM.

Homework 5 - Linear Transformations and Change of Basis - Due: November 21, 2026 (10%)

Problems from Chapter 8. Topics: linear maps between vector spaces, matrix representations, similar matrices, and Jordan normal form (intro).

Final Exam - Due: December 15, 2026 (20%)

Comprehensive final covering all course material. 2-hour exam held in the registrar-assigned room. One 8.5x11 sheet of notes (both sides) allowed. Bring a scientific calculator.

Course Schedule Notes

Week 1 (Sep 8-12): Vectors, dot products, matrix basics
Week 2 (Sep 15-19): Gaussian elimination, LU decomposition
Week 3 (Sep 22-26): Four fundamental subspaces, rank
Week 4 (Sep 29-Oct 3): Orthogonality, projections, Gram-Schmidt
Week 5 (Oct 6-10): QR factorization, least squares
Week 6 (Oct 13-17): Review + MIDTERM EXAM (Oct 16)
Week 7 (Oct 20-24): Determinants, eigenvalues intro
Week 8 (Oct 27-31): Diagonalization, spectral theorem
Week 9 (Nov 3-7): Singular Value Decomposition
Week 10 (Nov 10-14): SVD applications, pseudoinverse
Week 11 (Nov 17-21): PCA, dimensionality reduction
Week 12 (Nov 24-28): Linear transformations, change of basis [Wed-Fri only]
Week 13 (Dec 1-5): Numerical methods, iterative solvers
Week 14 (Dec 8-11): Review and exam preparation
FINAL EXAM: December 15, 2026

Grading Policy

Homework (5 sets): 50%

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Midterm Exam: 20%

Computational Project: 10%

Final Exam: 20%

Late policy: 10% penalty per day late. No submissions accepted after 3 days.

Grade scale: A 93+, A- 90, B+ 87, B 83, B- 80, C+ 77, C 73, C- 70, D 60, F <60.